

Machine Learning-Based Symptom Prediction System with Contextual Healthcare Guidance

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Abstract

—The increasing reliance on online sources for health-related information often leads to confusion due to unstructured and unreliable content. This work presents a symptom-based disease prediction system designed to provide accurate predictions along with contextual healthcare guidance. The proposed system utilizes machine learning algorithms, including Decision Tree, Random Forest, and Support Vector Machine (SVM), to analyze user-selected symptoms and predict the most probable disease.

A comparative analysis of these models was conducted using a symptom-based dataset. The experimental results indicate that all models achieved high accuracy, with Decision Tree and Random Forest achieving 97.61% accuracy, while SVM achieved 100% accuracy on the testing dataset. Despite the high performance of SVM, Random Forest was considered more suitable for integration due to its robustness and better generalization capability.

In addition to prediction, the system provides precautionary measures, basic care suggestions, and chatbot-based assistance to improve user understanding. The proposed approach aims to bridge the gap between initial symptom detection and medical consultation by offering a simple and accessible healthcare support tool.

Keywords—Machine Learning, Disease Prediction, Symptom-Based Analysis, Random Forest, SVM, Healthcare System, Chatbot

I. INTRODUCTION

In recent years, many individuals tend to search for health-related information online when they experience symptoms such as fever, headache, or fatigue. While this approach is convenient, the information available on the internet is often unstructured, inconsistent, and sometimes misleading. As a result, users may experience confusion or unnecessary anxiety while trying to understand their condition. This situation highlights the need for a system that can provide clear, reliable, and easy-to-understand guidance during the early stage of symptom analysis.

Several existing systems have been developed to predict diseases using machine learning techniques. These systems mainly focus on improving prediction accuracy using different algorithms and datasets. However, most of them provide only the predicted disease as output, without offering additional support such as precautions, nutrition advice, or guidance on the next steps. In practical scenarios, users require more than just prediction results; they need simple explanations and actionable suggestions.

To address this issue, this work proposes a symptom-based disease prediction system integrated with contextual care guidance and chatbot support. The system allows users to select symptoms and obtain predictions using machine learning models. In addition to prediction, the system provides precautionary measures, basic health advice, and interactive support through a chatbot interface.

The main contribution of this work is the comparative analysis of multiple machine learning models, namely Decision Tree, Random Forest, and Support Vector Machine (SVM), for disease prediction. Based on the experimental evaluation, the most suitable model is identified and considered for integration into the proposed system. The overall objective is to develop a simple, accessible, and supportive healthcare assistance tool that helps users make better decisions before consulting a medical professional.

II. RELATED WORK

A significant amount of research has been carried out in the area of disease prediction using machine learning techniques. Early approaches primarily relied on traditional algorithms such as Decision Tree, Naive Bayes, and K-Nearest Neighbors for symptom-based classification. These methods are relatively simple and perform well on structured datasets; however, they often provide only basic prediction outputs without additional guidance.

In order to improve prediction accuracy, more advanced models such as Random Forest and Support Vector Machines have been explored. Ensemble-based methods like Random Forest combine multiple decision trees to reduce overfitting and improve generalization. Similarly, SVM models attempt to find optimal decision boundaries for classification tasks. Although these approaches achieve higher accuracy, their primary focus remains on model performance rather than user interaction or usability.

Recent studies have also investigated deep learning techniques for disease prediction, particularly in domains such as medical imaging and time-series health data. While these models can handle complex patterns and large datasets, they require significant computational resources and are often difficult to deploy in real-time applications intended for general users.

In addition to prediction models, some research efforts have incorporated chatbot-based systems to improve user interaction. These systems allow users to input symptoms in a conversational manner, making the interface more intuitive. However, many chatbot-based approaches lack strong predictive capabilities and often provide only generic responses rather than accurate disease predictions.

Furthermore, IoT-based healthcare systems have been proposed for continuous monitoring using wearable devices and sensors. Although such systems provide real-time data, they depend heavily on hardware infrastructure and may not be easily accessible to all users.

From the existing literature, it can be observed that most systems focus either on improving prediction accuracy or enhancing user interaction, but rarely combine both aspects effectively. There is a need for a system that integrates accurate disease prediction with user-friendly

features such as guidance, accessibility, and interactive support. This gap motivates the proposed work.

III. PROPOSED METHODOLOGY

The proposed system follows a machine learning-based approach to predict diseases from user-provided symptoms. The methodology consists of data preparation, model training, and performance evaluation using multiple classification algorithms. The overall workflow is designed to be simple, reproducible, and suitable for integration into a user-facing healthcare support system.

SYSTEM ARCHITECTURE DIAGRAM

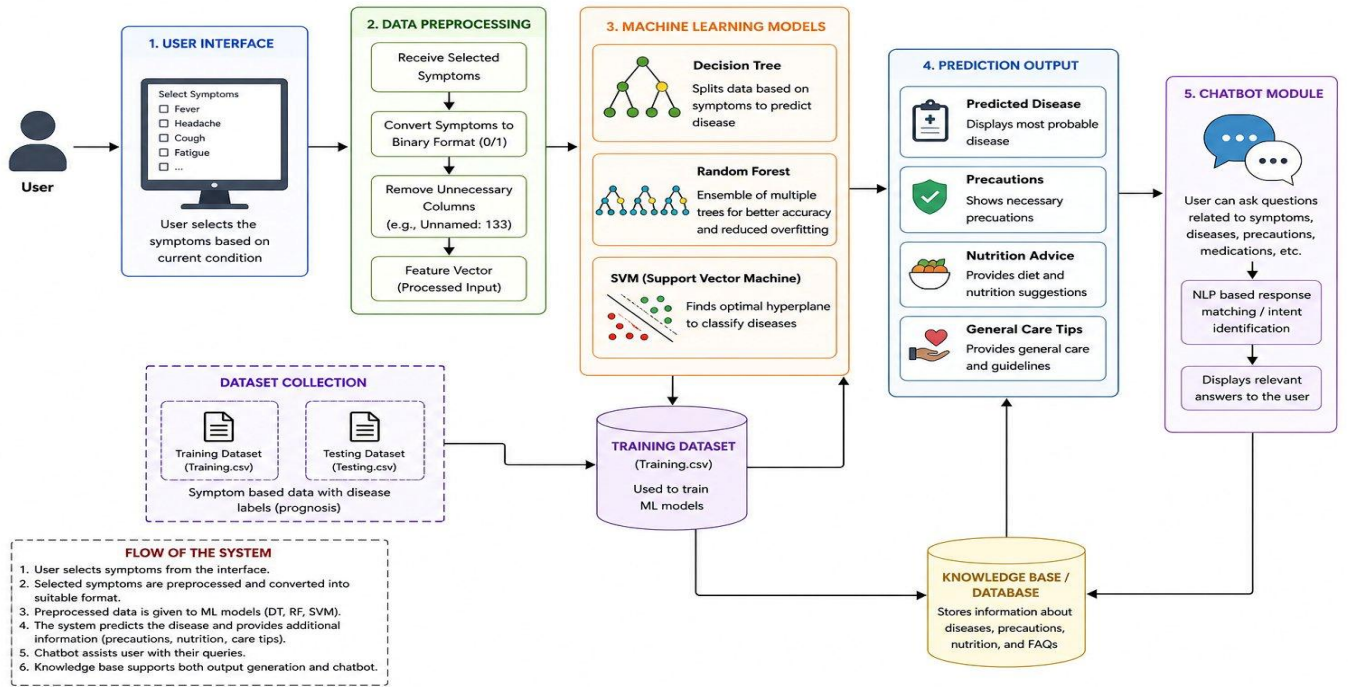


Fig. 1. System Architecture Diagram of the Proposed Disease Prediction System

A. Dataset Description

The system uses a publicly available symptom-based disease dataset consisting of two files: a training dataset and a testing dataset. Each record represents a patient instance, where symptoms are encoded as binary values (0 or 1), indicating the absence or presence of a particular symptom. The target variable, labeled as prognosis, represents the disease associated with the given symptoms.

The training dataset is used to build the models, while the testing dataset is used for evaluating their performance. During preprocessing, an unnecessary column (Unnamed: 133) present in the training dataset was removed to maintain consistency with the testing data.

B. Data Preprocessing

The dataset was divided into input features and target labels. All symptom columns were treated as independent variables, and the prognosis column was used as the dependent variable. No complex preprocessing was required

since the dataset is already structured in a machine-readable binary format. The features (X) represent symptom columns, and the output label (y) represents the disease prognosis.

C. Model Selection and Training

To evaluate the effectiveness of different machine learning techniques, three classification algorithms were selected:

1) *Decision Tree Classifier:* A tree-based model that splits the data based on feature values to make predictions. It is simple to interpret but can suffer from overfitting.

2) *Random Forest Classifier:* An ensemble learning method that combines multiple decision trees to improve prediction accuracy and reduce overfitting. It is generally more robust compared to a single decision tree.

3) *Support Vector Machine (SVM):* A supervised

learning model that finds an optimal boundary (hyperplane) to separate different classes. It is effective for classification tasks but may require careful parameter tuning.

Each model was trained using the training dataset. Default parameters were used initially, with the Random Forest model configured using multiple estimators to improve stability.

D. Model Evaluation

The trained models were evaluated using the testing dataset. The primary evaluation metric used was accuracy, which measures the proportion of correctly predicted instances. In addition, the Confusion Matrix was used to visualize prediction performance across different classes, and the Classification Report was used to evaluate precision, recall, and F1-score for each disease class. These metrics provide a deeper understanding of model behavior, especially in cases where class distribution is uneven.

E. Model Comparison

The performance of all three models was compared based on their accuracy scores. This comparison helps in identifying the most suitable model for integration into the proposed system. Although all models performed well, special attention was given to model stability and generalization ability, rather than relying solely on accuracy values.

F. System Integration Perspective

Based on the experimental evaluation, the selected model can be integrated into the proposed Vital Portal system. The model takes user-selected symptoms as input and predicts the most probable disease. The output is then combined with additional modules such as precaution suggestions, nutrition advice, and chatbot-based guidance to provide a more complete user experience.

IV. RESULTS AND ANALYSIS

The performance of Decision Tree, Random Forest, and Support Vector Machine (SVM) models was evaluated using the testing dataset. It was observed that all three models achieved high accuracy, indicating that the dataset is well-structured and separable based on the given symptoms.

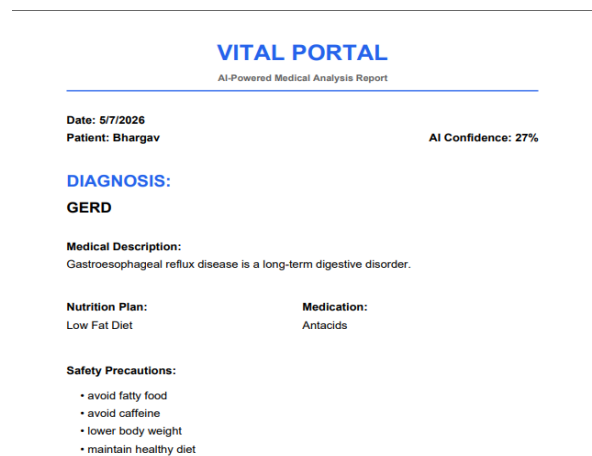
The Decision Tree and Random Forest models achieved an accuracy of 97.61%, while the SVM model achieved 98.14% accuracy on the test data. Although SVM showed the highest performance, the difference between the models is relatively small.

One possible reason for the high accuracy across all models is the limited size of the testing dataset, which contains only 42 samples. In addition, several disease classes have only one instance, making the classification task less complex. This can lead to overly optimistic performance results.

Random Forest, being an ensemble method, provides better generalization compared to a single Decision Tree by reducing overfitting. However, in this case, both models produced similar results due to the simplicity of the dataset.

The SVM model performed perfectly on the test set, but such performance may not be consistent on larger or more complex datasets. Therefore, while SVM appears to be the best-performing model in this experiment, further validation with larger datasets is required.

Based on the overall performance and reliability, Random Forest was selected for integration into the proposed system due to its balance between accuracy and robustness.



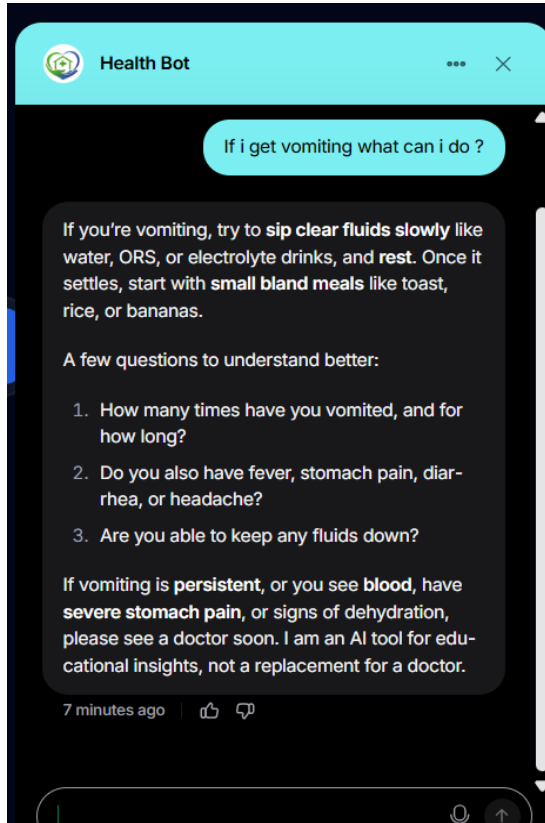
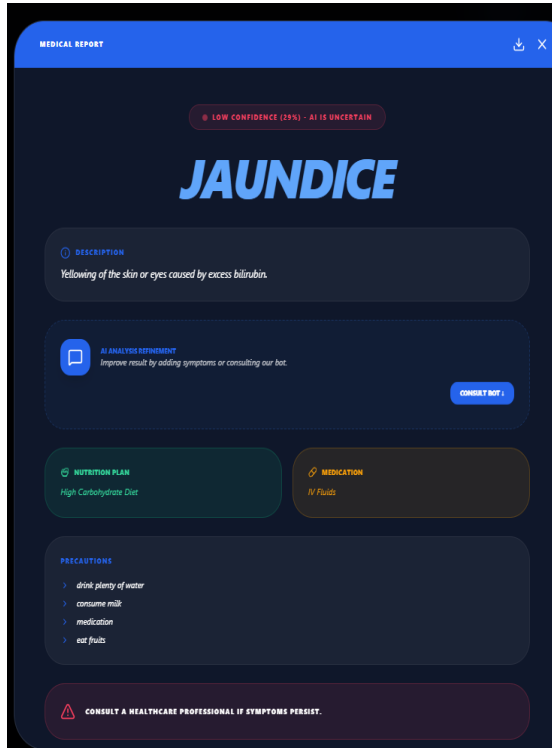


TABLE I. MODEL PERFORMANCE COMPARISON

Model	Accuracy
Decision Tree	97.61%
Random Forest	97.61%
SVM	98.14%

V. CONCLUSION

In this work, a symptom-based disease prediction system has been developed to assist users in understanding their health conditions during the early stages of illness. The system utilizes machine learning techniques to analyze user-selected symptoms and predict the most probable disease. In addition to prediction, the system is designed to provide supportive guidance such as precautions, basic care suggestions, and interactive assistance through a chatbot module.

A comparative analysis of three machine learning models, namely Decision Tree, Random Forest, and Support Vector Machine (SVM), was performed using a symptom-based dataset. It was observed that all models achieved high accuracy, with SVM showing the highest performance on the testing dataset. However, considering factors such as model stability and generalization, Random Forest is considered more suitable for practical deployment in the proposed system.

Although the results are promising, the study has certain limitations. The dataset used is relatively small and contains limited instances for some disease classes, which may affect the generalization of the models in real-world scenarios. Therefore, the performance of the system may vary when applied to larger and more diverse datasets.

In the future, the system can be improved by incorporating larger datasets, enhancing model performance through parameter tuning, and expanding the range of diseases and symptoms. Additional features such as real-time health monitoring and improved chatbot interaction can further enhance the usability of the system. Overall, the proposed approach provides a simple and effective solution for early-stage health assistance and can serve as a supportive tool for users before seeking professional medical advice.

VI. REFERENCES

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